Part 1

Problem 1:

Part 1

$$K = \begin{bmatrix} \alpha_x & s & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix}$$

α_x = focal length x-axis
α_y = focal length y-axis
s = skew
x_0, y_0 = principal points

Part 2:

3×4 Projective Camera = 12

12 − 1 = 11

In general, it has 5 parameters, rotation - 3 DOF, & Translation = 3

$\therefore 5 + 3 + 3 = 11$

Rectangular Pixels removes 1 parameter, so

$4 + 3 + 3 = 10$

Part 3:

$P = [M | P_4]$

Internal & External Parameters:

$P = KR[I | -C] = M[I | M^{-1}P_4]$

K = internal parameter
R = external parameter

Part 4:

Image of the point at ∞:

$D = (d^T, 0)^T$

$\therefore X = PD = [M | P_4](d^T, 0)^T$

Problem 2:

Part 1:

$PC = 0$

$x = \det([P_2, P_3, P_4])$
$z = \det([P_1, P_2, P_4])$
$y = -\det([P_1, P_3, P_4])$
$t = -\det([P_1, P_2, P_3])$

Part 2:

$S_x V_x = P_{xw} = [P_1, P_2, P_3, P_4][1,0,0,0]^T = P_1$
$S_y V_y = P_{yw} = [P_1, P_2, P_3, P_4][0,1,0,0]^T = P_2$
$S_z V_z = P_{zw} = [P_1, P_2, P_3, P_4][0,0,1,0]^T = P_3$

Part 3

Projection = $[x, y, 0]$

$$\begin{bmatrix} x \\ y \\ 0 \end{bmatrix} = \begin{bmatrix} P_1^T \\ P_2^T \\ P_3^T \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 0 \end{bmatrix}$$

Part 4:

The geometric meaning of the last row of P is that it encodes the plane. It describes the camera-centred plane where the image is the line at ∞.

Figure 1: Problems 1 and 2 process and computation

Part 2

SIFT (Scale Invariant Feature Transform) is a technique for extracting distinctive, scale-invariant features from images. These features are valuable for many computer vision applications, including image matching and object detection. The SIFT algorithm is composed of several major stages.

1. Scale-space extrema detection:
Potential interest points that remain consistent under changes in scale and orientation are first identified. This is done by detecting regions with significant intensity variation using the difference-of-Gaussians (DoG) function.

2. Keypoint localization:
Each point is refined by fitting a mathematical model to determine its precise position and scale. Keypoints are selected based on measures that indicate their stability.

3. Orientation assignment:
Each keypoint is assigned one or more orientations based on local image gradient directions. This ensures that the keypoint representation remains unaffected by image rotation.

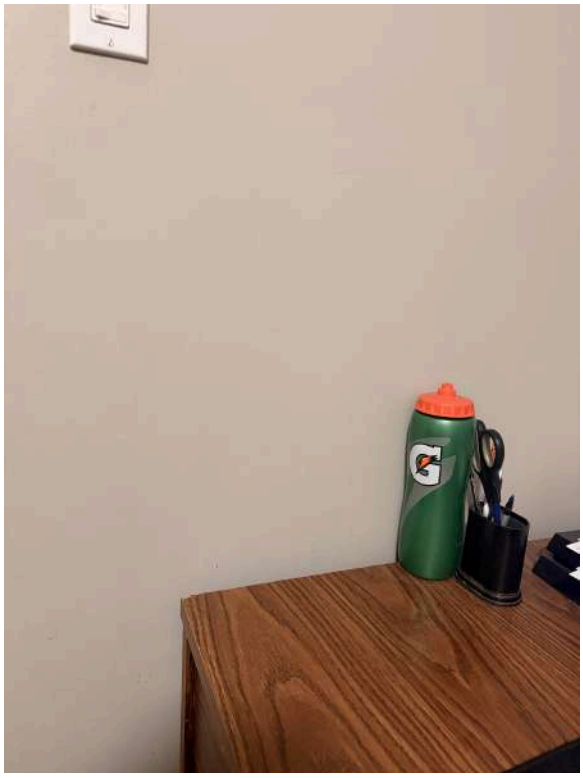
4. Keypoint descriptor creation:
Local image gradients around each keypoint are measured at the correct scale to understand how the intensity changes in that small region. These gradients are then transformed into a descriptor that is robust to changes in illumination and minor geometric distortions. The descriptor effectively captures the unique appearance around the keypoint.

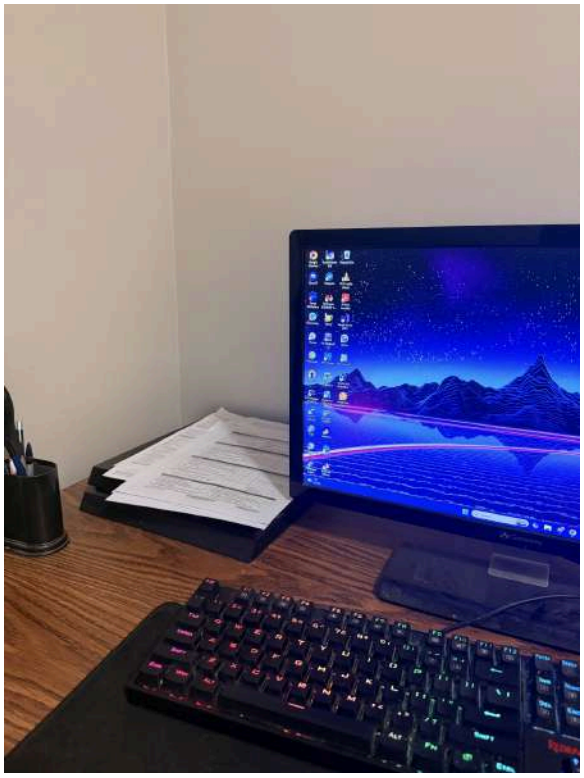
For image matching, SIFT descriptors extracted from reference images are stored in a database. When a new image is processed, its features are compared to those in the database using Euclidean distance to identify similar descriptors. This enables reliable matching across different viewpoints of the same object or scene.

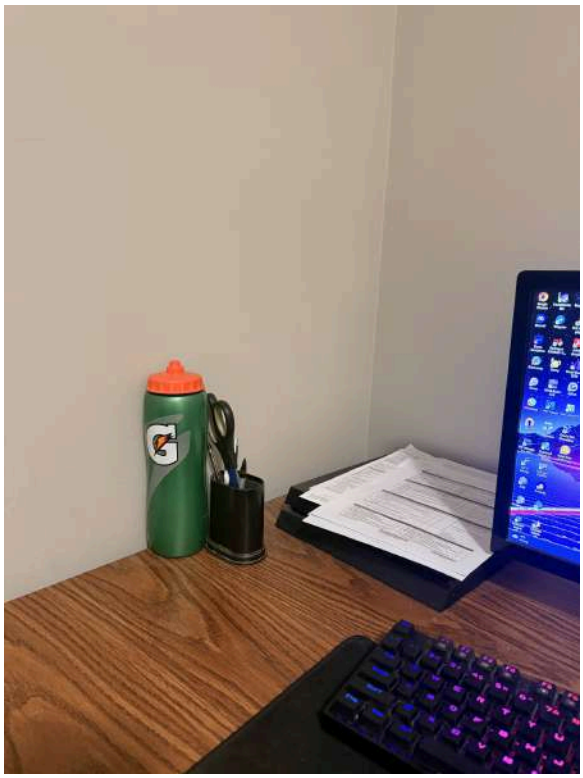
For image stitching, SIFT is used to detect distinctive points in each image. When two images overlap, SIFT finds pairs of points that match between them. These matching points help the system calculate how the images relate to each other, specifically, how much one image needs to be shifted, rotated, or slightly scaled to line up correctly with the other.

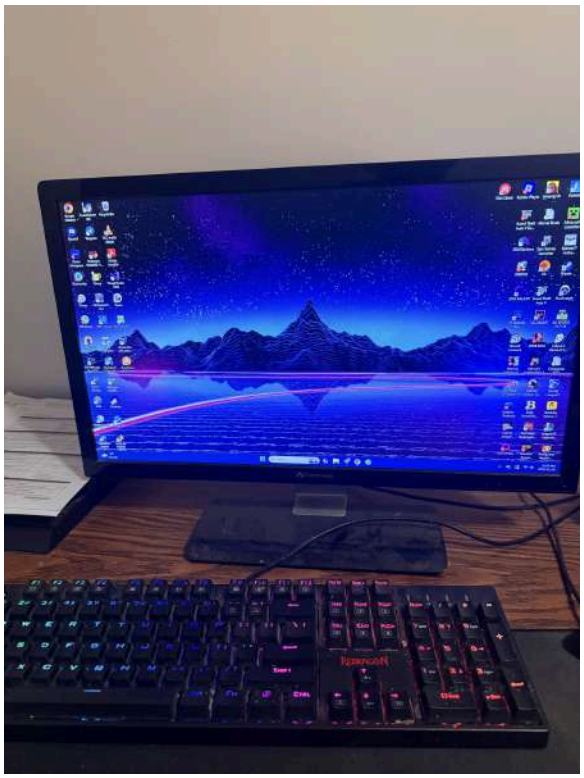
Overall, SIFT feature matching provides a reliable way to find correspondences between images, while the stitching pipeline aligns and merges them into a coherent panorama. This combination has become a standard approach in computer vision applications such as panorama creation, 3D reconstruction, and object recognition.

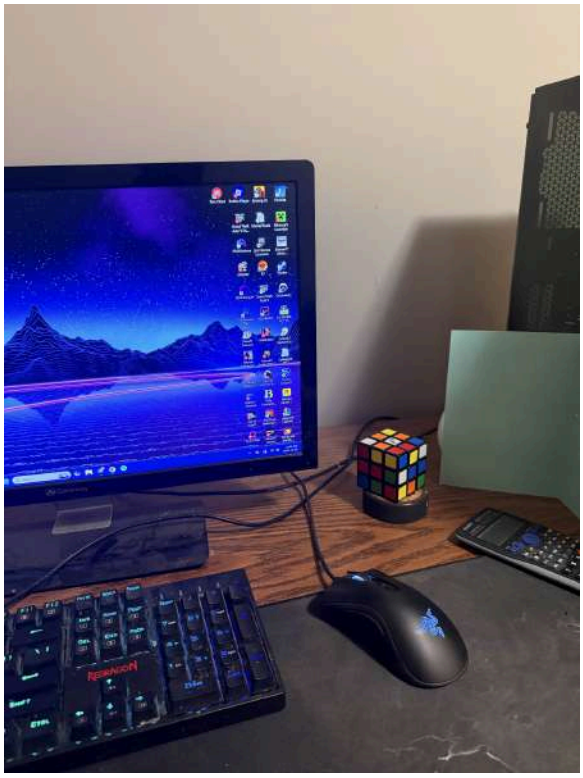
My output has looked relatively good, and the algorithm did a valid job of recognizing my images and stitching them together. The only issue was that the algorithm did not provide me with an output image, so I only had the grid version of it. This version was not straight and looked bent.



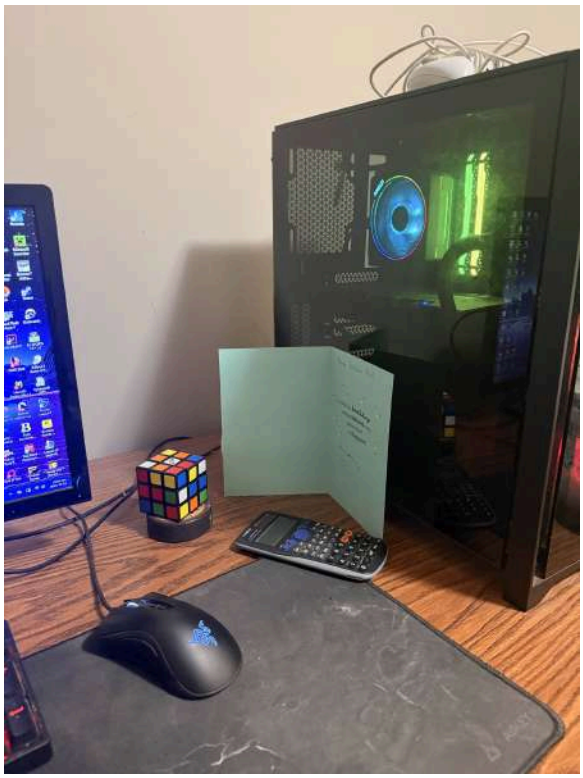














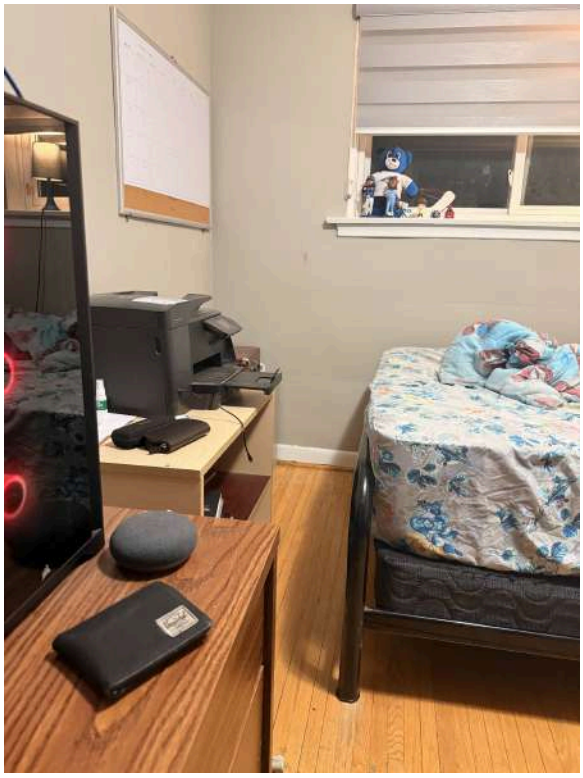




Figure 2: Input images and Output image of the panorama

Part 3

As for the project, my group are almost done with the coding base. We are getting successful outputs, and we're just fine-tuning it so it can be perfect. This week, we will be working on the progress report, as it is due by the end of the week.